

10_Database_Architecture.md

Part 3 — Security, Performance & Governance

Database Security Principles

The database must guarantee:

- Confidentiality
 - Integrity
 - Availability
 - Scalability
 - Auditability
-

Authentication Strategy

Authentication Provider

- Clerk

Authorization

- Role-Based Access Control (RBAC)

Roles

- Guest
 - User
 - Admin
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Authorization Matrix

Resource	Guest	User	Admin
Profile	✗	✓ Own	✓ All
Portfolio	✗	✓ Own	✓ All
Transactions	✗	✓ Own	✓ All
AI Conversations	✗	✓ Own	✓ All
Competitions	View	Join	Manage

Resource	Guest	User	Admin
Users	✗	✗	✓

Data Integrity Rules

- Every table must have a Primary Key.
- Foreign Keys are mandatory.
- No duplicate business records.
- Required fields cannot be NULL.
- Use transactions for multi-table operations.

Cascade Rules

Delete User

- Portfolio → Delete
- Holdings → Delete
- Transactions → Archive
- Watchlist → Delete
- Notifications → Delete
- AI Conversations → Archive

Delete Stock

Not allowed if referenced by:

- Holdings
- Transactions
- Watchlists

Instead:

```
```text id="w62j9f" is_active = false
```

```

```

```
Audit Logging
```

```
Every important action should be logged.
```

```
Tracked Events
```

```
- Login
```

- Logout
- Buy Order
- Sell Order
- Profile Update
- Password Change
- Competition Join
- Admin Actions

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## # Backup Strategy

### Daily Backup

- Full Database

### Hourly Backup

- Incremental Changes

### Retention

- Daily: 30 Days
- Weekly: 12 Weeks
- Monthly: 12 Months

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## # Recovery Strategy

### Recovery Objectives

- Automatic Backup Verification
- Point-in-Time Recovery
- Disaster Recovery Plan

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## # Performance Strategy

### Optimization Rules

- Index frequently queried columns.
- Avoid unnecessary joins.
- Use pagination.
- Cache market data.
- Archive historical records.

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## # Scalability Strategy

## Current Target

- 10,000 Users

## Future Target

- 1,000,000+ Users

## Scaling Plan

- Read Replicas
- Connection Pooling
- Redis Cache
- Partition Historical Data
- Background Jobs

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## # Data Retention Policy

Data	Retention
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Users	Until Account Deletion
Transactions	Permanent
AI Conversations	Configurable
Notifications	90 Days
Audit Logs	1 Year

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## # Monitoring

### Track

- Query Performance
- Slow Queries
- Failed Transactions
- Connection Usage
- Storage Growth
- Backup Status

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## # Database Governance

### Rules

- No direct production edits.
- Schema changes through migrations only.
- Every migration reviewed before deployment.

- Rollback plan required.
- Documentation updated after every schema change.

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## # Versioning

### Schema Version

```
```text id="h2vh4m"
v1.0.0
```

Migration Format

```
```text id="w55ndn" YYYYMMDD_description.sql
```

### Example

```
```text id="7t6gnt"
20260625_create_users_table.sql
```

Database Review Checklist

Before Release

- Tables normalized
- Foreign keys validated
- Indexes created
- Constraints verified
- Security reviewed
- Backup tested
- Performance tested
- Migration tested
- Documentation updated

Related Documents

- 08_Information_Architecture.md
- 09_Design_System.md
- 11_API_Architecture.md
- 12_Engineering_Standards.md

Approval Status

Version: 1.0.0

Status: Draft

End of Document